Inventory Management System for B2B SaaS

Overview

You're joining a team building "StockFlow" - a B2B inventory management platform. Small businesses use it to track products across multiple warehouses and manage supplier relationships.

Time Allocation

- **Take-Home Portion**: 90 minutes maximum
- **Live Discussion**: 30-45 minutes (scheduled separately)

Introduction & Assumptions

This document addresses three key areas of the StockFlow system:

- Debugging and improving an existing API
- Designing a scalable database schema
- Implementing a low-stock alert API

Given incomplete requirements, reasonable assumptions have been made and explicitly documented.

Part 1: Code Review & Debugging

A previous intern wrote this API endpoint for adding new products. **Something is wrong** - the code compiles but doesn't work as expected in production.

```
@app.route('/api/products', methods=['POST'])
def create_product():
    data = request.json

    # Create new product
    product = Product(
        name=data['name'],
        sku=data['sku'],
        price=data['price'],
        warehouse_id=data['warehouse_id']
    )

    db.session.add(product)
    db.session.commit()
```

```
# Update inventory count
inventory = Inventory(
    product_id=product.id,
    warehouse_id=data['warehouse_id'],
    quantity=data['initial_quantity']
)

db.session.add(inventory)
db.session.commit()

return {"message": "Product created", "product_id": product.id}
```

Your Tasks:

1. **Identify Issues**: List all problems you see with this code (technical and business logic)
2. **Explain Impact**: For each issue, explain what could go wrong in production
3. **Provide Fixes**: Write the corrected version with explanations

Additional Context (you may need to ask for more):

- Products can exist in multiple warehouses
- SKUs must be unique across the platform
- Price can be decimal values
- Some fields might be optional

1.1 Issues Identified

A. Validation Issues

- No validation for required fields (***name, sku, price, etc.***)
- No type validation (e.g., price should be decimal, quantity integer)

B. Data Integrity Issues

- SKU uniqueness not enforced
- No check if ***warehouse_id*** exists

C. Transaction Issues

- Two separate commits results in risk of partial writes

D. Business Logic Issues

- Assumes product belongs to only one warehouse
- Doesn’t handle optional fields

E. Reliability Issues

- No exception handling
- No logging
- No retry safety (idempotency)

1.2 Impact Analysis

- **Duplicate SKU** → breaks inventory tracking, ordering, and reporting
- **Partial writes** → product created but inventory missing → inconsistent system
- **Invalid inputs** → corrupt data affecting analytics and downstream systems
- **No error handling** → system crashes or silent failures in production

1.3 Fixed Implementation

```
from flask import request, jsonify
from sqlalchemy.exc import IntegrityError
from decimal import Decimal

@app.route('/api/products', methods=['POST'])
def create_product():
    data = request.json

    # Validation
    required_fields = ['name', 'sku', 'price', 'warehouse_id', 'initial_quantity']
    for field in required_fields:
        if field not in data:
            return jsonify({"error": f"{field} is required"}), 400

    try:
        price = Decimal(str(data['price']))
        quantity = int(data['initial_quantity'])
    except:
        return jsonify({"error": "Invalid price or quantity"}), 400

    try:
        with db.session.begin():

            # Check duplicate SKU
            existing = Product.query.filter_by(sku=data['sku']).first()
            if existing:
                return jsonify({"error": "SKU already exists"}), 400

            product = Product(
                name=data['name'],
                sku=data['sku'],
                price=price
            )
```

```
        db.session.add(product)
        db.session.flush()

        inventory = Inventory(
            product_id=product.id,
            warehouse_id=data['warehouse_id'],
            quantity=quantity
        )
        db.session.add(inventory)

    return jsonify({
        "message": "Product created",
        "product_id": product.id
    }), 201

except IntegrityError:
    db.session.rollback()
    return jsonify({"error": "Database constraint error"}), 400

except Exception as e:
    db.session.rollback()
    return jsonify({"error": "Internal server error"}), 500
```

1.4 Failure Scenarios & Mitigation

Scenario	Mitigation
Duplicate SKU	DB unique constraint + validation
Partial DB writes	Transaction (*db.session.begin*)
Invalid input	Input validation
Retry request	(Future) idempotency key
DB failure	Rollback + error handling

Part 2: Database Design (25 minutes)

Based on the requirements below, design a database schema. **Note**: These requirements are intentionally incomplete - you should identify what's missing.

Given Requirements:

- Companies can have multiple warehouses
- Products can be stored in multiple warehouses with different quantities

- Track when inventory levels change
- Suppliers provide products to companies
- Some products might be "bundles" containing other products

Your Tasks:

1. **Design Schema**: Create tables with columns, data types, and relationships
2. **Identify Gaps**: List questions you'd ask the product team about missing requirements
3. **Explain Decisions**: Justify your design choices (indexes, constraints, etc.)

Format: Use any notation (SQL DDL, ERD, text description, etc.)

2.1 Schema

Company

- id (PK)
- name
- created_at
- updated_at

Warehouse

- id (PK)
- company_id (FK → Company.id)
- name
- created_at
- updated_at

Product

- id (PK)
- sku (UNIQUE)
- name
- price
- created_at
- updated_at

Inventory (Junction Table)

- id (PK)
- product_id (FK → Product.id)
- warehouse_id (FK → Warehouse.id)
- quantity (CHECK ≥ 0, DEFAULT 0)
- created_at
- Updated_at

Constraints:

- Unique (product_id, warehouse_id)

Supplier

- id (PK)
- name
- contact_email
- created_at
- updated_at

ProductSupplier (Many-to-Many)

- product_id (FK → Product.id)
- supplier_id (FK → Supplier.id)

Constraints:

- Composite PK (product_id, supplier_id)

InventoryHistory (Audit Table)

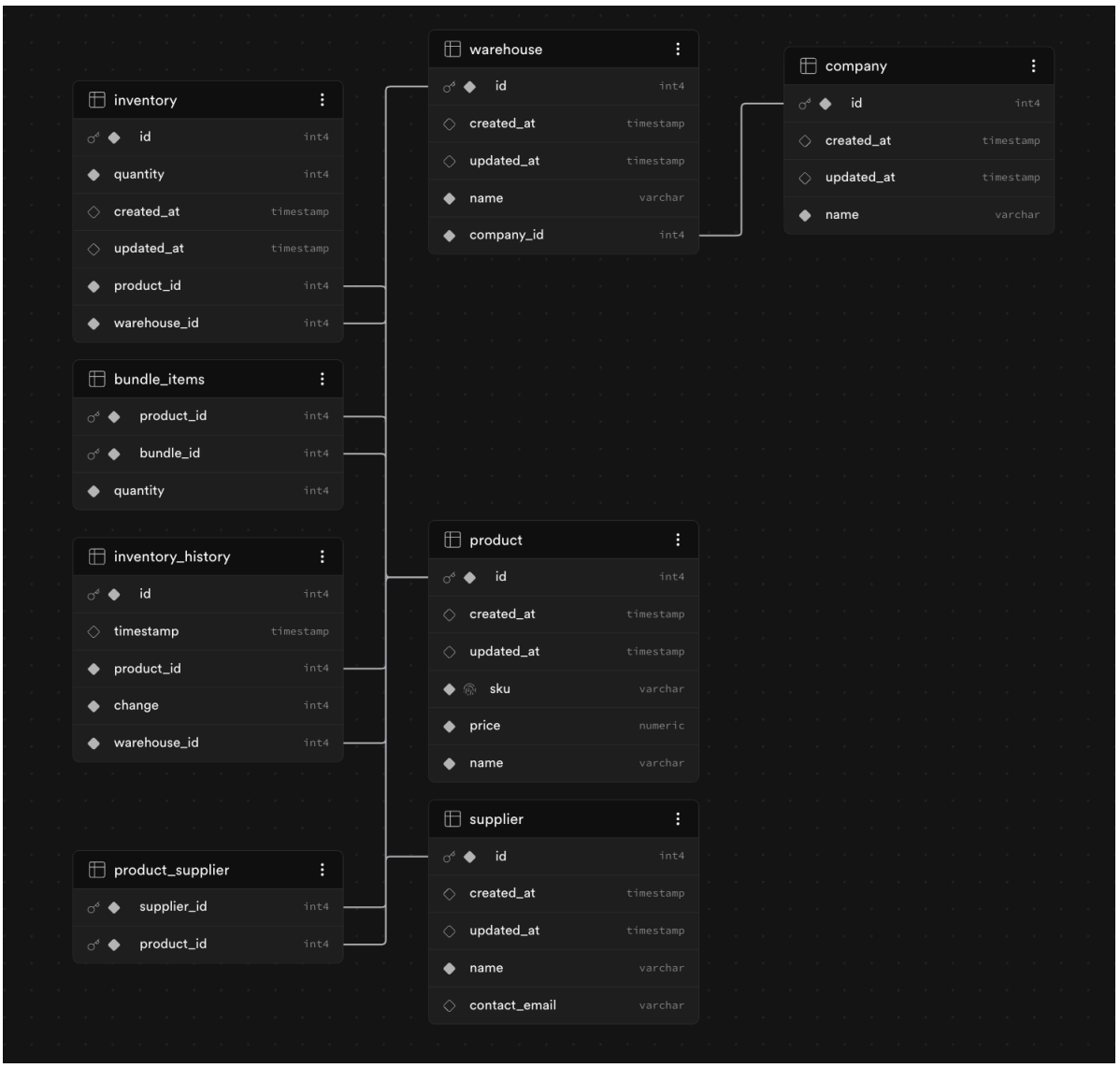
- id (PK)
- product_id (FK → Product.id)
- warehouse_id (FK → Warehouse.id)
- change
- timestamp

BundleItems

- bundle_id (FK → Product.id)
- product_id (FK → Product.id)
- quantity

Constraints:

- Composite PK (bundle_id, product_id)



2.2 Design Decisions

- **Inventory as junction table** → supports multi-warehouse storage
- **InventoryHistory** → audit + analytics support
- **ProductSupplier** → many-to-many flexibility
- **BundleItems** → composite product modeling

2.3 Constraints & Indexing Strategy

Constraints

- SKU is **globally unique**
- Inventory unique per (product_id, warehouse_id)
- Foreign key constraints enforced
- Quantity ≥ 0 (business rule assumption)

Indexing Strategy

Indexes are designed based on actual query patterns:

- **Product lookup**

  UNIQUE INDEX (sku)

- **Company → Warehouse queries**

  INDEX (company_id)

- **Inventory joins**

  INDEX (product_id, warehouse_id)

- **Low-stock filtering**

  INDEX (warehouse_id, quantity)

- **Supplier lookup**

  INDEX (product_id) in product_supplier

- **Inventory history tracking**

  INDEX (product_id, timestamp DESC)

Part 3: API Implementation (35 minutes)

Implement an endpoint that returns low-stock alerts for a company.

Business Rules (discovered through previous questions):

- Low stock threshold varies by product type
- Only alert for products with recent sales activity
- Must handle multiple warehouses per company
- Include supplier information for reordering

Endpoint Specification:

GET /api/companies/{company_id}/alerts/low-stock

Expected Response Format:

```
{
  "alerts": [
    {
      "product_id": 123,
      "product_name": "Widget A",
      "sku": "WID-001",
      "warehouse_id": 456,
```

```
      "warehouse_name": "Main Warehouse",
      "current_stock": 5,
      "threshold": 20,
      "days_until_stockout": 12,
      "supplier": {
        "id": 789,
        "name": "Supplier Corp",
        "contact_email": "orders@supplier.com"
      }
    }
  ],
  "total_alerts": 1
}
```


Your Tasks:

1. **Write Implementation**: Use any language/framework (Python/Flask, Node.js/Express, etc.)
2. **Handle Edge Cases**: Consider what could go wrong
3. **Explain Approach**: Add comments explaining your logic

Hints: You'll need to make assumptions about the database schema and business logic. Document these assumptions.

Submission Instructions

1. **Create a document** with your responses to all three parts
2. **Include reasoning** for each decision you made
3. **List assumptions** you had to make due to incomplete requirements
4. **Submit within 90 minutes** of receiving this case study
5. **Be prepared** to walk through your solutions in the live session

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Live Session Topics (Preview)

During our video call, we'll discuss:

- Your debugging approach and thought process
- Database design trade-offs and scalability considerations
- How you'd handle edge cases in your API implementation
- Questions about missing requirements and how you'd gather more info
- Alternative approaches you considered

Evaluation Criteria

Technical Skills:

- Code quality and best practices
- Database design principles
- Understanding of API design
- Problem-solving approach

Communication:

- Ability to identify and ask about ambiguities
- Clear explanation of technical decisions
- Professional collaboration style

Business Understanding:

- Recognition of real-world constraints
- Consideration of user experience
- Scalability and maintenance thinking

Note: This is designed to assess your current skill level and learning potential. We don't expect perfect solutions - we're more interested in your thought process and ability to work with incomplete information.